

Network Security – Part 3: Cheat Sheets & Career

OSI Model

L	Name	Example	Security
7	Application	HTTP, DNS	TLS
6	Presentation	TLS/SSL	Encryption
5	Session	NetBIOS	Timeout
4	Transport	TCP, UDP	Port filter
3	Network	IP, OSPF	ACLs
2	Data Link	Ethernet, Wi-Fi	WPA3
1	Physical	Fiber	Physical locks

Ports & Protocols

Port	Proto	Use	Secure?
22	SSH	Remote shell	Yes
23	Telnet	Remote shell	Never
53	DNS	Name resolution	DoH 853
80	HTTP	Web	→ 443
443	HTTPS	Secure web	Yes
25/587	SMTP	Email	TLS
110/143	POP3/IMAP	Email	995/993 TLS
3389	RDP	Remote desktop	VPN+MFA
161	SNMP	Monitoring	v3 only
389/636	LDAP	Directory	636 LDAPS

Subnetting Quick Ref

192.168.1.0/24 = 254 hosts | /25=126 | /26=62 | /27=30 | /28=14 | /29=6 | /30=2
Private: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

Wireshark Filters

Filter	Finds
dns	DNS
http	HTTP
tcp.port==443	HTTPS
tcp.flags.syn==1 and tcp.flags.ack==0	New connections
icmp	Ping

Nmap Cheat Sheet

Command	Purpose
nmap -sn 192.168.1.0/24	Ping sweep
nmap -sV 192.168.1.1	Service versions

nmap -p 22,80,443 192.168.1.0/24	Specific ports
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Carrier / Telecom Security Career

Path: NOC Analyst → SOC Analyst → Network Security Engineer

Certs: Cisco CCNA, ISC2 CC, Security+

Skills: BGP, OSPF, MPLS, firewalls, DDoS, RADIUS/TACACS+, NetFlow

Portfolio: 3 Wireshark pcaps + 1 hardening report

Glossary

Term	Meaning
OSI	7-layer network model
TCP/IP	Core internet protocols
DNS	Name to IP translation
TLS	Encrypts web traffic
VPN	Encrypted tunnel
Firewall	Filters traffic
SIEM	SOC monitoring tool
NOC	Network Operations Center
SOC	Security Operations Center
PCAP	Packet capture file

12-Week Tracker

Week	Hours	Labs	Notes
1	_____	_____	
2	_____	_____	
3	_____	_____	
4	_____	_____	
5	_____	_____	
6	_____	_____	
7	_____	_____	
8	_____	_____	
9	_____	_____	
10	_____	_____	
11	_____	_____	
12	_____	_____	